Fiscal Year:	FY 2004	**Task Last Updated:**	FY 05/12/2004
PI Name:	Evans, Michael L Ph.D.		
Project Title:	Analysis of the Interaction of Environmental Signal Transduction Pathways using Gravi-compensation		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Organismal biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	43210	**Congressional District:**	15
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	99-HEDS-02
Start Date:	06/15/2000	**End Date:**	12/31/2003
No. of Post Docs:	1	**No. of PhD Degrees:**	2
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	Evans, Michael		
COI Name (Institution):	Ishikawa, Hideo (The Ohio State University)		
Grant/Contract No.:	NAG2-1411		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The primary task is to use newly designed software and hardware to dissect the interaction of signaling and responses for two major root response systems -- phototropism and gravitropism. Our goals include characterizing the sensing and response sites for phototropism and determining the extent to which the sensing and response machinery of phototropism overlap with comparable response systems for gravitropism. The key objective was to characterize the nature of light sensing by roots and the extent to which the response mechanism for light sensing and gravity sensing overlap. A thorough understanding of how plants sense and interact with their environment will improve our ability to manipulate and improve plant growth and promote efforts to increase the ability of plants to survive and be productive under less than ideal conditions.		

Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Although there has been intensive investigation of root responses to gravity, it is clear that roots are sensitive to a variety of other environmental stimuli such as light, touch, and temperature. During the evolution of the signal transduction pathways for these response system it is likely that common second messenger systems have been incorporated and that there is overlap or cross-talk between the pathways. Our objectives are to examine the interaction of phototropism and gravitropism using two recently designed devices. The first is a black box fitted with infrared viewing ports, stimulus irradiation Light Emitting Diode (LED) and fiber optics. The device allows us to utilize our plant growth imaging software on roots growing in the dark with uninterrupted recording following unilateral illumination. The second device is a closed feedback system that couples our image analysis software with root manipulation motors that allow constant maintenance of specified root regions at user-selected angles. This system allows automated maintenance of the root apex in a vertical orientation during responses to unilateral illumination, thus avoiding the modifying effect of gravity that normally accompanies phototropism. Using this device, we are measuring "pure" phototropism in the absence of gravity-induced growth asymmetry so that we can conduct a detailed study of root phototropism including an investigation of the interaction of phototropism transduction/response pathways with the response of roots to gravity to gain insight into the nature of both responses.
Task Progress:	Our efforts during this final year of funding were focused on two areas: 1) an investigation of patterns of auxin distribution from columella cells to lateral root cap cells in Arabidopsis using a Green Fluorescent Protein (GFP)-based auxin sensor, and 2) use of our previously described seedling manipulation technology (ROTATO: feedback device for plant manipulation during video imaging) to analyze the motor phases of root gravitropism in maize. For the work on auxin distribution patterns we collaborated with the laboratory of Klaus Palme at the University of Freiburg in Germany. The results from the GFP study indicate that auxin levels are elevated in columella cells and that gravistimulation causes rapid asymmetric redistribution of auxin from the columella toward lateral root cap cells (LRC) along the lower side of the root and then back along the lower side toward the elongation zone. This pattern of gravistimulated auxin movement is as predicted by the classical Cholodny-Went hypothesis. Movement from columella to LRC was primarily regulated by the auxin efflux carrier while movement from LRC toward the elongation zone involved both the auxin influx carrier and the efflux carrier. The analysis of the gravitropic motor response in maize roots revealed previously unknown features of the response. The key findings were: 1) the motor response occurs as two phases of downward curvature (a rapid Phase I and a slower Phase II) separated by a period of straightening (S phase); 2) these phases occur independently of the changing strength of the stimulus as the root approaches vertical (as revealed by ROTATO analysis during which curvature can be analyzed while maintaining the root tip at the original angle of stimulation); 3) the S phase of the response is strongest at low angles of stimulation and disappears at (constant) high angles of stimulation; and 4) phase II curvature does not occur at low (< about 30 degrees) angles of stimulation; under these conditions all curvature is accounted for by phase I. These findings are particularly significant in view of previous findings on this project. We now know that there are two gravity sensing regions in the root (cap and distal elongation zone), two motor regions in the root (distal elongation zone and central elongation zone), and two phases to the curvature response (Phase I and Phase II). Further analysis is needed to determine how these features of gravitropism are interrelated. In addition, the phenotypes of the wide variety of known gravitropic mutants need to be described in terms of these newly recognized key features of gravitropism.
Bibliography Type:	Description: (Last Updated: 03/08/2019)
Abstracts for Journals and Proceedings	Ishikawa H, Natori E, Evans ML. "An examination of long term motor activity in maize root gravitropism under conditions of constant stimulus input." Annual meeting, American Society for Gravitational and Space Biology, Huntsville, Alabama. November 12-16 (2003) Gravitational and Space Biology Bulletin 17: pg 52 , Nov-2003